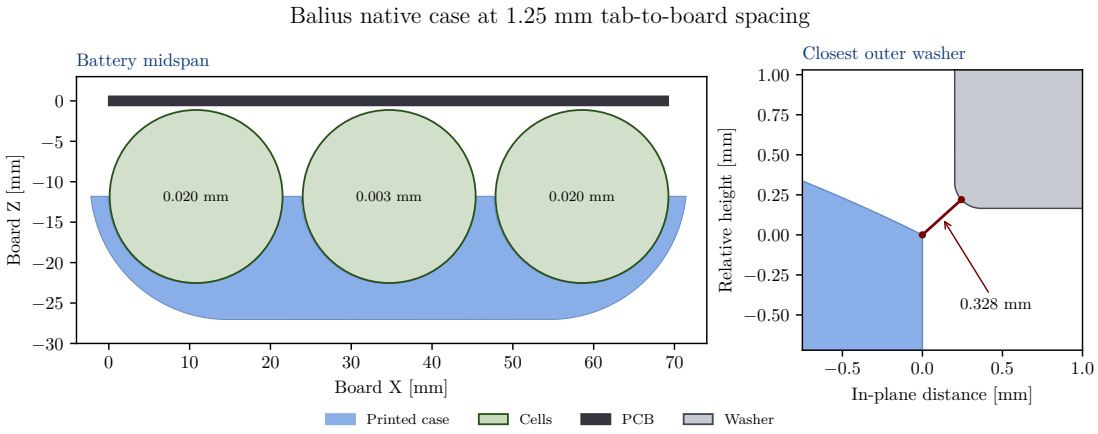
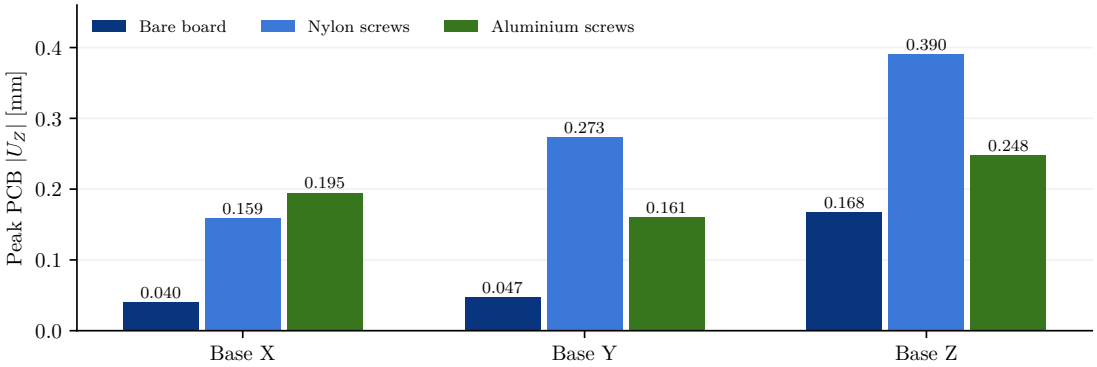
The three-pocket Balius case fits at 1.25 mm stand-off



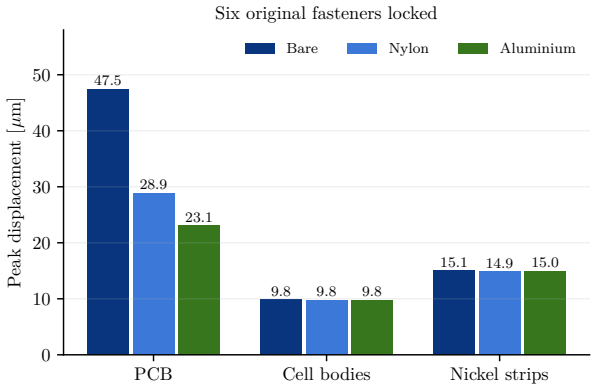
The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. The rigid left cell reaches its pocket after **0.020 mm toward the middle cell** or **0.165 mm away from it**.

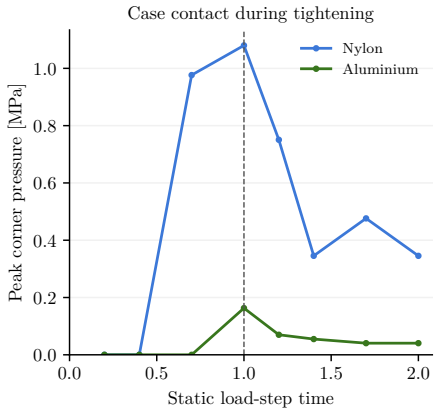
The snug case increases board travel in the seeded clips



During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

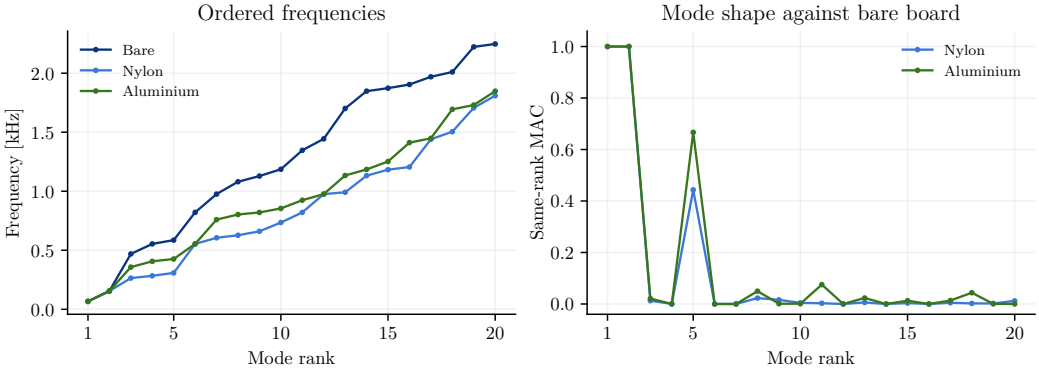
Both snug mounts reduce PCB movement during tightening





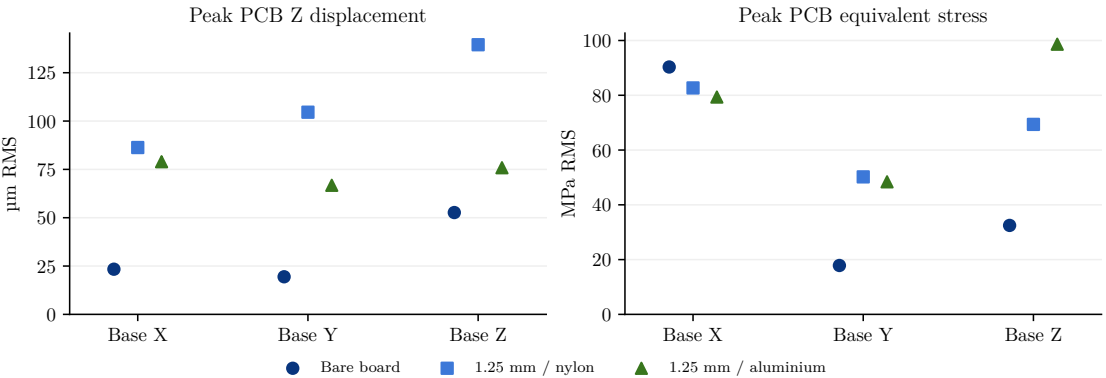
At bolt lock, peak PCB displacement fell from **47.46 μm bare** to **28.91 with nylon screws** and **23.09 with aluminium screws**. All three nylon meshes put the peak at the same corner, with a spread of 0.055 μm. Only the middle cell touched the case.

The first unpreloaded mode stays at 66.61 Hz



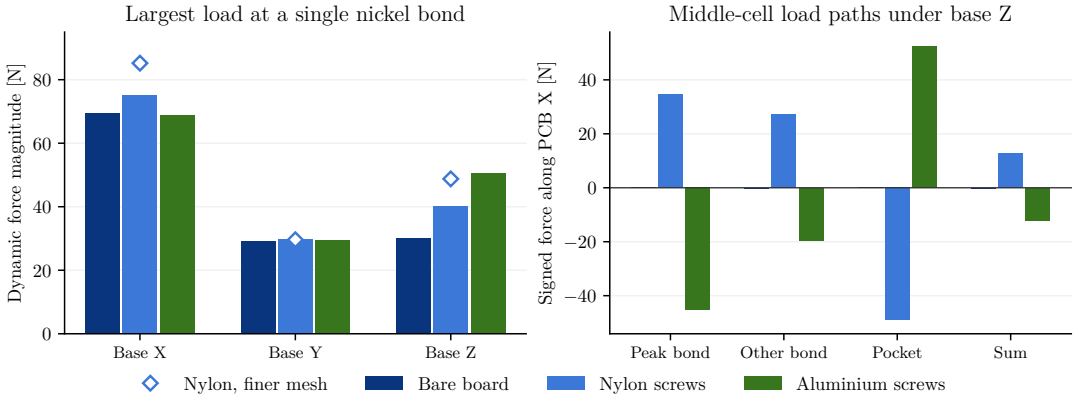
The first two mode shapes closely match the bare board. At mode 20 the shapes differ, so comparing frequencies by rank alone would give a misleading shift. These are the refined default-contact controls before bolt tightening. Paired movies.

Aluminium cuts board travel relative to nylon but raises base-Z stress



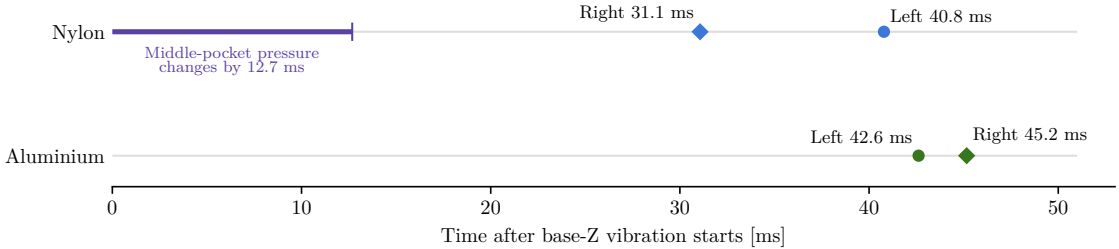
Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

Board-normal vibration loads the middle cell's nickel bond



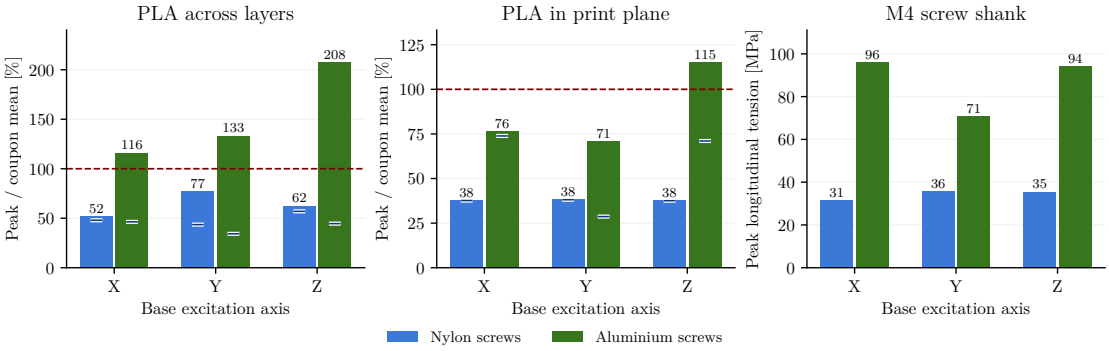
The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The finer nylon mesh gave **48.8 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories at the installed contact state. The diamond uses independent integration on the finer native modes.

The first outer cell to reach the case changes with the screws



The native base-Z history puts the **right cell first at 31.1 ms with nylon** and the **left cell first at 42.6 ms with aluminium**. Independent integration gives the same order. These times assume contact stays as it was at bolt lock. With nylon, the middle pocket already needs to redistribute its load by 12.7 ms. Impact forces remain unresolved.

The printed tab exceeds its coupon reference with aluminium screws



On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.